Sabrina Abram, PhD

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Summary

Computational cognitive neuroscientist studying adaptive behavior in humans. My research combines reinforcement learning models with behavioral and neuropsychological experiments to understand **how agency shapes human behavior and how behavior changes when that agency is shared with other systems**. This research has informed human-in-the-loop systems and extends to fundamental questions of agency and alignment in AI systems.

Education

University of California, Berkeley, Postdoctoral Fellow — 2022–present
Departments of Psychology and Neuroscience

Stanford University, Visiting Scholar — 2019–2020
Department of Mechanical Engineering

Simon Fraser University, PhD — 2015–2021
Department of Engineering Science

Cleveland State University, BSc Summa Cum Laude — 2011–2015
Department of Mechanical Engineering; Division I Athlete

Experience

10+ years of hypothesis-driven and data-driven science

Cognition and Action Lab, University of California, Berkeley
Advisor: Rich Ivry, Focus: Agency and reinforcement learning

- Identified distinct cognitive biases in decision making driven by agency in action selection and action execution
- Developed reinforcement learning models that found dissociable neural mechanisms of agency in decision making
- Managed teams across Berkeley, Yale, and Carnegie Mellon, and presented results at 4 international conferences

Biomechatronics Lab, Stanford University
Advisor: Steve Collins, Focus: Human-in-the-loop optimization

- Analyzed longitudinal datasets of human-exoskeleton interaction to reveal explore-exploit strategies in humans
- Identified computational mechanisms and behavioral markers of learning during human-in-the-loop optimization
- Published findings in *Current Biology*, with an invited dispatch article discussing the broader impact of the work

Locomotion Lab, Simon Fraser University
Advisor: Max Donelan, Focus: Energy optimization in human locomotion

- Designed and built a robotic device to determine whether humans optimize energy as they learn new movements
- Developed computational models of energy optimization to identify how humans learn in high-dimensional spaces
- Awarded Canada's top doctoral fellowship for research and leadership; presented at 10 international conferences

Publications

In Preparation

1. **Abram SJ**, Bolo Dave L, Tsay JS, Wang T, Collins AGE, McDougle SD, and Ivry RB. Agency in action selection and action execution produce distinct biases in decision making.
2. **Abram SJ**, Tsay JS, Wang T, Li JJ, Collins AGE, McDougle SD, and Ivry RB. Differential roles of the cerebellum and basal ganglia in agency-dependent reinforcement learning.
3. Sukumar S, **Abram SJ**, Chirino-Pérez A, and Ivry RB. Cerebellar degeneration impairs volitional category switching in semantic fluency.

4. Cisneros E, Bol K, Collins AGE, **Abram SJ**, Ivry RB, and Tsay JS. No evidence for the role of the cerebellum in value-based decision making.
5. **Abram SJ**, Ortega J, Wang T, Whitney D, and Ivry RB. Cerebellar mechanisms of dynamic emotion tracking.

Journal Articles, Book Chapters, and Refereed Conference Proceedings

6. **Abram SJ**, Chirino-Pérez A, Wang T, and Ivry RB (Under Review). Cerebellar contributions to human social cognition. In *Advances in Neurobiology*. Springer.
7. **Abram SJ**, Tsay JS, Wang T, Collins AGE, McDougle SD, and Ivry RB (2025). Agency in action selection and action execution produce distinct biases in decision making. *Proceedings of the 6th RLDM Conference*.
8. Cisneros E, Bol K, Collins AGE, **Abram SJ**, Ivry RB, and Tsay JS (2025). No evidence for the role of the cerebellum in value-based decision making. *Proceedings of the 6th RLDM Conference*.
9. Chen Y, **Abram SJ**, Ivry RB*, and Tsay JS* (2025). Indirect feedback hinders explicit sensorimotor adaptation. *Proceedings of the Royal Society B: Biological Sciences*.
10. **Abram SJ***, Tsay JS*, Yosef H, Reisman DS, and Kim HE (2025). The detrimental effect of stroke on motor adaptation. *Neurorehabilitation and Neural Repair*.
11. Wang T, Avraham G, Tsay JS, **Abram SJ**, and Ivry RB (2024). Perturbation variability does not influence implicit sensorimotor adaptation. *PLOS Computational Biology*.
12. van Mastrigt NM, Tsay JS, Wang T, Avraham G, **Abram SJ**, van der Kooij K, Smeets JB, and Ivry RB (2023). Implicit reward-based motor learning. *Experimental Brain Research*.
13. **Abram SJ**, Poggensee KL, Sánchez N, Simha SN, Finley JM, Collins SH, and Donelan JM (2022). General variability leads to specific adaptation toward optimal movement policies. *Current Biology*.
Featured in Current Biology Dispatch: Motor control: In constant pursuit of optimality.
14. Simha SN, Wong JD, Selinger JC, **Abram SJ**, and Donelan JM (2021). Increasing the gradient of energetic cost does not initiate adaptation in human walking. *Journal of Neurophysiology*.
15. **Abram SJ**, Selinger JC, and Donelan JM (2019). Energy optimization is a major objective in the real-time control of step width in human walking. *Journal of Biomechanics*.
16. Kashiri N, Abate A, **Abram SJ**, et al. (2018). An overview on principles for energy efficient robot locomotion. *Frontiers in Robotics and AI*.

Skills

Technical Skills: Programming (Matlab, Python, R); Data analysis (mixed-effects models, permutation testing, bootstrapping); Reinforcement learning (parameter estimation and recovery, model simulation and comparison)

Science Communication: 14 international conference presentations spanning cognitive neuroscience, psychology, artificial intelligence, machine learning, and robotics, including invited talks and award-winning presentations

Leadership: Formally supervised over 10 students (2 Honors) at Berkeley and Simon Fraser University; mentored students in STEM and Women in Science programs at University of British Columbia and Canadian Science Expo

Service

Reviewer: Brain; IEEE Transactions on Neural Systems & Rehabilitation Engineering; Journal of Biomechanics; Journal of Experimental Biology; Neuroscience; Communications Psychology; Experimental Brain Research, NPJ

Committee Member: Society for the Neural Control of Movement; Parkinson's Disease Research Symposium

Captain: NCAA Division I Women's Soccer; achieved program records in scoring, shutouts, and team GPA

Scholarships and Awards (Selected)

New Investigator Scholarship, Society for the Neural Control of Movement 2024
Michael Smith Foreign Study Supplement, NSERC Canada 2019–2020
Vanier Canada Graduate Scholarship ($150,000), NSERC Canada 2016–2019
NCAA Division I Women's Soccer Scholarship, Cleveland State University 2011–2015